THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A lightning conductor protects buildings by: 1

(a) attracting rain **(b)** passing the charge safely to the earth
(c) insulating the roof **(d)** creating sound

2. Friction always acts: 1

(a) upwards
(b) at right angles always
(c) along the motion
(d) opposite to the direction of motion or tendency of motion

3. Which is a disadvantage of friction? 1

(a) lighting a match **(b)** wearing out of shoe soles
(c) walking on road **(d)** holding a pen

4. When a glass rod is rubbed with silk, the rod becomes: 1

(a) negatively charged **(b)** a magnet
(c) positively charged **(d)** neutral

5. When two forces on a body are equal and opposite, the forces are: 1

(a) balanced **(b)** frictionless **(c)** unbalanced **(d)** magnetic

6. Which is an advantage of friction? 1

(a) machines wear out **(b)** tyres wear
(c) we can walk without slipping **(d)** engines get heated

7. A magnet attracts: 1

(a) plastic **(b)** wood **(c)** paper **(d)** iron

8. Force can be measured using a: 1

(a) stopwatch **(b)** thermometer **(c)** ammeter **(d)** spring balance

9. A boy pushes a heavy box but it does not move. The friction acting is: 1

(a) rolling friction **(b)** static friction equal to the push
(c) zero **(d)** sliding friction larger than push

10. Two like magnetic poles: 1

(a) repel **(b)** have no effect **(c)** neutralise **(d)** attract

INFINITY

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the use of a lightning conductor? 2

12. What is friction? Why does it arise? 2

13. What do you observe when a plastic scale is rubbed on dry hair and brought near small bits of paper? Why? 2

14. Why is a spring balance used to measure force? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three advantages of friction. 3

16. Differentiate between mass and weight (three points). 3

17. What is muscular force? Give three examples. 3

OR

Give three disadvantages of friction. 3

18. Explain why a heavy box needs a larger force to start moving than to keep it moving. 3

OR

Explain how the force of air resistance (drag) slows down a moving object. Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain electrostatic force with the example of a rubbed balloon. Describe charging, attraction and repulsion. 5

OR

Explain how a magnet exerts a non-contact force. Describe the forces between poles and explain how to make a magnet float. 5

20. How does a lightning conductor work? Why is lightning dangerous? State four safety rules during a thunderstorm. 5

OR

Explain, with examples, how pressing, pulling, stretching and twisting are applications of force. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND